

Grade 8
Unit 2
Lesson 4 Practice

Name _____
Date _____
Period _____

1. Find the square root of each of the following.

$$\sqrt{81} =$$

$$\sqrt{-81} =$$

2. Find the cube root of each.

$$\sqrt[3]{64} =$$

$$\sqrt[3]{-64} =$$

3. Why is there a non-real root for $\sqrt{-81}$, however there is a real root for the $\sqrt[3]{-64}$?

Consider $\sqrt{95}$, an irrational number.

4. Between which two perfect squares does $\sqrt{95}$ lie?

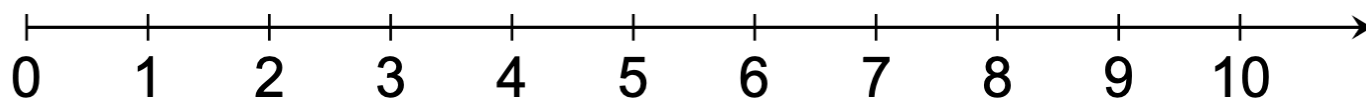
5. Between which two integers is the value of $\sqrt{95}$?

Now consider $\sqrt[3]{141}$, another irrational number.

6. Between which two perfect cubes does $\sqrt[3]{141}$ lie?

7. Between which two integers is the value of $\sqrt[3]{141}$?

8. Plot and label the approximation of $\sqrt{95}$ and $\sqrt[3]{141}$ on the number line.



9. Lauren was trying to approximate the solution to $n^2 = 115$. She said that it lies between $\sqrt{100}$ and $\sqrt{121}$. Therefore, it lies between 4 and 5, however a little closer to 5. She approximated it to be about 4.7. Do you agree or disagree with her reasoning? Explain.

10. Approximate the solution to $n^2 = 115$ and then plot value on the number line below.

